Low-noise and boundary asymptotics for depolarizing-channel squashed entanglement

Abstract

For $\Lambda_p(X) = (1-p)X + p\ \mathrm{Tr}(X)I_2/2$, squashed entanglement with arbitrary finite quantum squashing outputs satisfies

$$E_{\mathrm{sq}}(\Lambda_p) = 1 - \frac{3p}{8}\log_2\frac{1}{p} + O(p) \quad (p \downarrow 0).$$

At $p = 2/3 - 4\delta/3$, its value is bounded above and below by positive constant multiples of δ^2. An explicit map from the canonical four-dimensional environment to a sixteen-dimensional output works for every input; its exact maximum is attained at the Bell input. Its cost is $J(r(\delta))$, with

$$J(r) = \frac{\kappa r^2}{2\ln 2} + O(r^3), \qquad \kappa = \frac{2059}{5600}\ln 29 - \frac{2171}{5800} > 0.$$

The remainders are uniform on the stated physical intervals.

Takeoka, Guha, and Wilde introduced channel squashed entanglement as a bound on two-way assisted quantum and private communication [5]. Discarding the squashing environment gives the standard upper bound of half the entanglement-assisted classical capacity. For the depolarizing channel this is

$$E_{\mathrm{sq}}(\Lambda_p) \le 1 - \tfrac{1}{2}H(1-3p/4, p/4, p/4, p/4),$$

where H is the base-two Shannon entropy. Its low-noise expansion already has the coefficient $3/8$ in the abstract. The new low-noise claim is the matching lower bound for arbitrary finite quantum squashing outputs. The boundary estimates and the explicit squashing map are stated separately below.

1 Definitions and the endpoint theorems

Every state and channel below acts on finite-dimensional complex Hilbert spaces. Density matrices may be singular. Use natural entropy $S_{\mathrm{nat}}(\rho) = -\operatorname{Tr}\rho\ln\rho$, with $0\ln 0 = 0$, and bit entropy $S_2 = S_{\mathrm{nat}}/\ln 2$. Conditional mutual information is

$$I_2(R:B\mid E) = S_2(RE) + S_2(BE) - S_2(E) - S_2(RBE), \qquad I_{\mathrm{nat}} = (\ln 2)I_2.$$

For a state,

$$E_{\mathrm{sq}}(\rho_{RB}) = \inf_{\substack{\omega_{RBE}\ \text{density} \\ \dim E<\infty,\ \mathrm{Tr}_E\,\omega=\rho_{RB}}} \frac{1}{2}I_2(R:B\mid E)_\omega.$$

Let $P_0 = I, P_1 = X, P_2 = Y, P_3 = Z$, and put

$$q_0 = 1 - \frac{3p}{4}, \quad q_1 = q_2 = q_3 = \frac{p}{4}, \qquad U_p|\psi\rangle = \sum_{i=0}^{3}\sqrt{q_i}\,P_i|\psi\rangle_B|i\rangle_{\mathsf{E}}.$$

For $0 \le p \le 1$ this is an isometry implementing Λ_p. The canonical environment E has dimension four. The channel value has the following order of quantifiers:

$$E_{\mathrm{sq}}(\Lambda_p) = \sup_{\substack{\dim R<\infty \\ |\psi\rangle_{RA}\ \text{unit}}} \frac{1}{2}\inf_{\substack{\dim E<\infty \\ \mathcal{S}:\mathsf{E}\to E\ \mathrm{CPTP}}} I_2(R:B\mid E)_{(\mathrm{id}_{RB}\otimes\mathcal{S})(\mathrm{id}_R\otimes U_p)|\psi\rangle\langle\psi|(\mathrm{id}_R\otimes U_p^\dagger)}. \tag{1.1}$$

All finite squashing output dimensions occur in the inner infimum. Write $|\Phi\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, $P = |\Phi\rangle\langle\Phi|$, and

$$\rho_\delta = \left(\frac{1}{2} + \delta\right) P + \left(\frac{1}{2} - \delta\right) \frac{I - P}{3}, \qquad p(\delta) = \frac{2}{3} - \frac{4\delta}{3}.$$

The normalized Choi state of $\Lambda_{p(\delta)}$ is ρ_δ.

Theorem 1.1. *There is $C_0 > 0$ such that, for every $0 < p \le 1/2$,*

$$\left| E_{\mathrm{sq}}(\Lambda_p) - 1 + \frac{3p}{8} \log_2 \frac{1}{p} \right| \le C_0 p. \tag{1.2}$$

There is $C_1 > 0$ such that, for every $0 \le \delta \le 21/512$,

$$\frac{\delta^2}{9 \ln 2} \le E_{\mathrm{sq}}(\Lambda_{p(\delta)}) \le C_1 \delta^2. \tag{1.3}$$

In particular $E_{\mathrm{sq}}(\Lambda_{p(\delta)}) = \Theta(\delta^2)$ as $\delta \downarrow 0$.

Theorem 1.2. *For every $0 \le \delta \le 21/512$ there is an explicitly defined CPTP map $\mathcal{S}_\delta : \mathsf{E} \to \mathbb{C}^{16}$ such that its half-CMI cost over all normalized pure reference/input states has a maximum attained at the Bell input, equal to*

$$J(r) = \frac{H(0,r) + H(r,0) - H(r,r) - H(0,0)}{2 \ln 2}, \qquad r = r(\delta) = \frac{320\delta}{\sqrt{324 + 21120\delta} + 18}, \tag{1.4}$$

$$H(x,y) = S_{\mathrm{nat}}((D_x \otimes D_y)\xi), \qquad \xi = \frac{7}{8}|v\rangle\langle v| + \frac{I_4}{32},$$

$$|v\rangle = \frac{2|00\rangle + |11\rangle}{\sqrt{5}}.$$

Here D_x is amplitude damping with Kraus operators $\mathrm{diag}(1, \sqrt{1-x})$ *and $\sqrt{x}|0\rangle\langle 1|$. There are finite constants $M_r, M_\delta \ge 0$ such that*

$$\left| J(r) - \frac{\kappa}{2 \ln 2} r^2 \right| \le M_r r^3 \qquad (0 \le r \le 1/4), \tag{1.5}$$

$$\left| J(r(\delta)) - \frac{3200\kappa}{81 \ln 2} \delta^2 \right| \le M_\delta \delta^3 \qquad (0 \le \delta \le 21/512). \tag{1.6}$$

The coefficient κ is the cost coefficient of this displayed map.

2 Finite extensions and the Bell input

Every finite extension of the output associated with an input in (1.1) is obtained by a finite channel on its purifying environment. To see this, purify the extension in an additional finite system G. Two purifications of the same output marginal are related by an isometry on the support of the purifying marginal. Extend its associated channel off that support by mapping the unused trace to a fixed output density. Conversely a channel on the environment supplies an extension with the same output marginal. Thus the inner infimum is exactly the state squashed entanglement of that input's output.

For this Pauli channel,

$$E_{\mathrm{sq}}(\Lambda_p) = E_{\mathrm{sq}}((\mathrm{id} \otimes \Lambda_p)P). \tag{2.1}$$

Here is the finite channel argument. Standard Bell measurement and Pauli correction simulate Λ_p from its Choi state: for the resource $(I \otimes P_i)P(I \otimes P_i)^\dagger$, teleportation implements $X \mapsto P_i X P_i$; their mixture with weights q_i implements Λ_p. This is a finite LOCC protocol [2, Appendix B, Proposition 26]. The pure input and its reference start locally with the sender. Adding them does not change the resource's squashed entanglement, and LOCC monotonicity [1] bounds the output value by that resource's value. Bell input gives the reverse inequality. The extension argument above identifies the resource infimum with the canonical-environment infimum. This proves (2.1) with the order in (1.1).

3 A uniform binary-square entropy bound

For positive, possibly subnormalized X, set $S_{\mathrm{sub}}(X) = -\operatorname{Tr} X \ln X$. For a binary orthogonal pinching Δ, define its natural entropy cost $C(X) = S_{\mathrm{sub}}(\Delta X) - S_{\mathrm{sub}}(X)$. Pinching preserves trace, so $C(\alpha X) = \alpha C(X)$.

Lemma 3.1. *Let H be any finite Hermitian binary-block matrix. Put*

$$t = \operatorname{Tr} H^2, \qquad u = \|PH(I-P)\|_{\mathrm{HS}}^2.$$

If $0 < u \le t/2$, then

$$C(H^2) \le u\,[\mathcal{F}(t/u) + 4], \qquad \mathcal{F}(m) = \operatorname{arcosh}(m/2)\sqrt{\frac{m+2}{m-2}}, \quad \mathcal{F}(2) = 2. \tag{3.1}$$

At $u = 0$, $C(H^2) = 0$. For $m \ge 2$,

$$\mathcal{F}(m) \le \ln m + \frac{2\ln m}{m-1} \le \ln m + 2. \tag{3.2}$$

There is no assumption on the signs of the diagonal blocks or on rank.

Proof. Write

$$H = \begin{pmatrix} A & Z^\dagger \\ Z & B \end{pmatrix}, \quad \rho = H^2, \quad D = \Delta\rho = D_0 \oplus D_1,$$

$$D_0 = A^2 + Z^\dagger Z, \qquad D_1 = B^2 + ZZ^\dagger, \qquad Y = ZA + BZ.$$

For $x > 0$ let

$$S_x = (D_1 + xI)^{-1/2} Y (D_0 + xI)^{-1/2}, \quad Q_x = S_x^\dagger S_x < I.$$

The strict inequality is the Schur complement of $\rho + xI > 0$. The determinant formula gives

$$\ln\det(D + xI) - \ln\det(\rho + xI) = -\operatorname{Tr}\ln(I - Q_x).$$

Integrate the scalar primitive $(x+\lambda)\ln(x+\lambda) - (x+\lambda)$ over the two spectra. Equal dimension and trace cancel the infinite endpoint. The finite endpoint is $\operatorname{Tr}\rho\ln\rho - \operatorname{Tr} D\ln D$. Zero eigenvalues produce integrable $\ln x$ terms, and the infinite tail is $O(x^{-2})$. Thus

$$C(\rho) = \mathcal{J}(Y) + N, \tag{3.3}$$

where

$$\mathcal{J}(V) = \int_0^\infty \operatorname{Tr} V^\dagger (D_1 + xI)^{-1} V (D_0 + xI)^{-1}\, dx,$$
$$N = \int_0^\infty \operatorname{Tr}[-\ln(I - Q_x) - Q_x]\, dx.$$

If $z \in \ker D_0$, then $Az = Zz = 0$; if $z \in \ker D_1$, then $Bz = Z^\dagger z = 0$. All subsequent inverses and logarithms can be restricted to these positive ranges without changing the integrals.

We first bound the nonlinear term. Let $H_0 = A \oplus B$ and $W = H - H_0$. For every $\varepsilon > 0$, expansion of a positive Hermitian square gives

$$D = H_0^2 + W^2 \le (1+\varepsilon)\rho + (2 + \varepsilon^{-1}) W^2. \tag{3.4}$$

Let $w_k = \sigma_k(Z)^2$ be in decreasing order. Split $W^2 = F_k + R_k$, keeping its largest $k-1$ singular modes in each sector. Then $\operatorname{rank} F_k \le 2(k-1)$ and $0 \le R_k \le w_k I$. For $w_k > 0$, put $r = x/w_k$, $\varepsilon = (\sqrt{1+r} - 1)^{-1}$, and $c = 1 - (1+r)^{-1/2}$. These satisfy $c(1+\varepsilon) = 1$ and $(1-c)x = c(2+\varepsilon^{-1})w_k$. Equation (3.4) implies

$$\rho + xI - c(D + xI) \ge -c(2 + \varepsilon^{-1}) F_k.$$

After congruence by $(D + xI)^{-1/2}$, the matrix $\left(\begin{smallmatrix} I & S_x^\dagger \\ S_x & I \end{smallmatrix}\right)$ is at least cI minus a positive matrix of rank at most $2(k-1)$. Its $(2k-1)$st smallest eigenvalue is therefore at least c. Its small eigenvalues are $1 - \sigma_j(S_x)$, with extra ones for unequal sectors. Consequently

$$q_{2k-1}(x), q_{2k}(x) \le \frac{w_k}{x + w_k}, \quad q_j(x) = \sigma_j(S_x)^2.$$

Extend the lists by zero; a pair with only one index decreases the sum. For $f_0(q) = -\ln(1-q) - q$, which is increasing and nonnegative,

$$\int_0^\infty f_0\!\left(\frac{w}{x+w}\right) dx = w \int_0^1 \frac{f_0(q)}{q^2}\, dq = w.$$

An antiderivative in the last integral is $-\ln(1-q) - f_0(q)/q$, with endpoint values 0 and 1. It follows that

$$N \le 2 \sum_k w_k = 2u. \tag{3.5}$$

Next let $P_0 = \sqrt{D_0}$, $P_1 = \sqrt{D_1}$, and $R_i = (D_i + xI)^{-1}$. The diagonal blocks A, B may be indefinite. Nevertheless inverse order and congruence give

$$AR_0A \le A(A^2 + xI)^{-1} A = A^2 (A^2 + xI)^{-1} \le D_0 (D_0 + xI)^{-1} = P_0 R_0 P_0.$$

The analogous inequality holds for B. Taking positive traces yields

$$\mathcal{J}(ZA) + \mathcal{J}(BZ) \le \mathcal{J}(ZP_0) + \mathcal{J}(P_1 Z).$$

The mixed form is the integral of the Hilbert–Schmidt inner product of BR_1Z and ZR_0A. Each of their integrated squared norms is at most u, because $B^2 \le D_1, A^2 \le D_0$ and $\int_0^\infty d/(d+x)^2\, dx = 1$ on positive support. Two applications of Cauchy–Schwarz give

$$\mathcal{J}(Y) \le \mathcal{J}(ZP_0) + \mathcal{J}(P_1 Z) + 2u. \tag{3.6}$$

In separate eigenbases of P_0, P_1, the self-term is

$$\sum_{b,a} |Z_{ba}|^2 (p_a^2 + q_b^2) \frac{\ln p_a^2 - \ln q_b^2}{p_a^2 - q_b^2}.$$

The equal-eigenvalue formula is its continuous limit. On Hilbert–Schmidt matrices let $X = \ln L_{D_1} - \ln R_{D_0}$. The commuting left and right multiplication operators give a spectral probability measure associated with $Z/\sqrt{u}$. For $y = |X|$, the last display is $u\,\mathbb{E}[y \coth(y/2)]$, taking value 2 at $y = 0$. The inverse contractions

$$Z D_0^{-1} Z^\dagger \le I, \qquad Z^\dagger D_1^{-1} Z \le I$$

imply $\langle Z, e^X Z\rangle \le \operatorname{Tr} D_1$ and $\langle Z, e^{-X} Z\rangle \le \operatorname{Tr} D_0$. Thus

$$\mathbb{E} \cosh y \le \frac{t}{2u}.$$

With $r = \sinh^2(y/2)$, direct integration gives

$$y \coth(y/2) = 2 \int_0^1 \sqrt{\frac{1+r}{1+v^2 r}}\, dv.$$

For $\beta \in [0,1]$, $g_\beta(r) = \sqrt{(1+r)/(1+\beta r)}$ is increasing and concave:

$$g_\beta' = \frac{(1-\beta) g_\beta}{2(1+r)(1+\beta r)}, \qquad g_\beta'' = -\frac{(1-\beta)(1+3\beta+4\beta r) g_\beta}{4(1+r)^2(1+\beta r)^2}.$$

Since $\mathbb{E} r \le (t/u - 2)/4$, Jensen gives

$$\mathcal{J}(ZP_0) + \mathcal{J}(P_1 Z) \le u \mathcal{F}(t/u).$$

Combine this with (3.3), (3.5), and (3.6) to obtain (3.1).

Finally $y \coth(y/2)$ is increasing, since its derivative multiplied by $\sinh^2(y/2)$ is $(\sinh y - y)/2 \ge 0$. Use $\operatorname{arcosh}(m/2) \le \ln m$ and $\coth(\ln m/2) = 1 + 2/(m-1)$ to obtain the first bound in (3.2). The second follows from $\ln m \le m - 1$. □

For later use, if T is a rectangular amplitude whose input and output each have a binary decomposition, let

$$t_T = \|T\|_{\mathrm{HS}}^2, \qquad u_T = \|T_{01}\|_{\mathrm{HS}}^2 + \|T_{10}\|_{\mathrm{HS}}^2.$$

The Hermitian dilation $\left(\begin{smallmatrix} 0 & T \\ T^\dagger & 0 \end{smallmatrix}\right)$, grouped by the two binary labels, has budgets $t = 2t_T, u = u_T$. Its square has opposite Grams. When $u_T > 0$, Lemma 3.1 gives

$$C_{\mathrm{out}}(TT^\dagger) + C_{\mathrm{in}}(T^\dagger T) \le u_T[\mathcal{F}(2t_T/u_T) + 4]. \tag{3.7}$$

At $u_T = 0$, both costs are zero. The sectors may have unequal dimensions.

4 Four-image entropy sharing at low noise

Put $s = 3p/4$, $a = p/4 = s/3$, $b = 1 - s$, and $w = b + a = 1 - 2a$. A finite CPTP squashing map has a finite Stinespring isometry $\mathsf{E} \to E \otimes G$. Its four images $v_i = \mathrm{vec}(M_i)$ are arbitrary orthonormal vectors:

$$M_i \in \mathbb{C}^{d_E \times d_G}, \qquad \mathrm{Tr}\, M_i^\dagger M_j = \delta_{ij}.$$

The pure squashed Bell extension is

$$|\Psi\rangle = \sqrt{b}\,|\Phi\rangle v_0 + \sqrt{a}\sum_{j=1}^{3}(I \otimes P_j)|\Phi\rangle\, v_j.$$

Its AE/BG rectangular amplitude, and its AG/BE amplitude, are

$$T_E = \frac{1}{\sqrt{2}}\left[\sqrt{b}\, I \otimes M_0 + \sqrt{a}\sum_{j=1}^{3} P_j^{\mathsf{T}} \otimes M_j\right], \tag{4.1}$$
$$T_G = \frac{1}{\sqrt{2}}\left[\sqrt{b}\, I \otimes M_0^{\mathsf{T}} + \sqrt{a}\sum_{j=1}^{3} P_j^{\mathsf{T}} \otimes M_j^{\mathsf{T}}\right].$$

Each has Hilbert–Schmidt norm one. Its left Gram is the named first marginal; its physical right marginal is the total transpose of the right Gram. Entropy and the pinching by real computational projectors are invariant under that transpose. The environment marginals are exactly

$$\rho_E = bM_0M_0^\dagger + a\sum_j M_jM_j^\dagger, \qquad \rho_G = \left(bM_0^\dagger M_0 + a\sum_j M_j^\dagger M_j\right)^{\mathsf{T}}.$$

Hence the half-CMI bit cost U satisfies

$$\begin{aligned} U &= \frac{1}{2}[S_2(AE) - S_2(E) + S_2(AG) - S_2(G)] \\ &= 1 - \frac{1}{4}J_4, \qquad J_4 = I_2(A:E) + I_2(A:G) + I_2(B:E) + I_2(B:G). \end{aligned} \tag{4.2}$$

The four-MI identity follows from purity and maximally mixed A, B; it uses both environment entropy subtractions.

For a spin/environment marginal, computational pinching gives the exact decomposition

$$I_2(A:E) = I_2(Z_A:E) + C_A(\rho_{AE})/\ln 2.$$

There are four such decompositions. In (4.1) the two off-spin blocks are $\sqrt{a/2}(M_1 + iM_2)$ and $\sqrt{a/2}(M_1 - iM_2)$, each of squared norm a. Each of the two rectangular dilations therefore has $t = 2, u = 2a$. Equation (3.7) bounds the sum of four coherence costs, in bits, by

$$\frac{4a}{\ln 2}[\mathcal{F}(1/a) + 4]. \tag{4.3}$$

Let $A_0 = \sqrt{b}M_0$, $V = \sqrt{a}M_3$. The two conditional E states after A pinching are

$$(A_0 \pm V)(A_0 \pm V)^\dagger + a(M_1 \pm iM_2)(M_1 \pm iM_2)^\dagger.$$

Each has trace $w + 2a = 1$. The coherent branch of weight w is the equal-prior ensemble

$$v_{\pm} = \frac{\sqrt{b}\, v_0 \pm \sqrt{a}\, v_3}{\sqrt{w}}.$$

Write χ_E, χ_G for its two reduced Holevo quantities. The XY branch has weight $2a$ and equal priors. For B pinching, the same coherent branch appears, while the XY states exchange. Convexity of mutual information at fixed classical prior and the binary information bound 1 give

$$\sum_{\text{four marginals}} I_2(Z : \text{environment}) \le 2w(\chi_E + \chi_G) + 8a. \tag{4.4}$$

The equality of priors in both branches justifies this convexity step.

Form a second rectangular matrix

$$L = \frac{1}{\sqrt{2}} \begin{pmatrix} A_0 & V \\ V & A_0 \end{pmatrix}.$$

It has squared norm w. A Hadamard on each binary copy label turns it into the blocks $(A_0 + V)/\sqrt{2}, (A_0 - V)/\sqrt{2}$, each of squared norm $w/2$. The block entropy formula gives

$$C(LL^\dagger) + C(L^\dagger L) = (\ln 2) w(\chi_E + \chi_G).$$

Its Hermitian dilation has budgets $t = 2w, u = a$. Therefore

$$(\ln 2) w(\chi_E + \chi_G) \le a[\mathcal{F}(2w/a) + 4].$$

Combine this with (4.2), (4.3), and (4.4). For every finite squashing map,

$$U \ge 1 - \frac{a}{\ln 2}\mathcal{F}(1/a) - \frac{a}{2\ln 2}\mathcal{F}(2w/a) - \frac{6a}{\ln 2} - 2a. \tag{4.5}$$

This bound holds throughout $0 < p \le 1$; both ratios are at least 4.

In the original p convention the natural loss in (4.5) is

$$\mathcal{L}(p) = \frac{p}{4}\mathcal{F}(4/p) + \frac{p}{8}\mathcal{F}((8-4p)/p) + (6 + 2\ln 2)\frac{p}{4}.$$

Using (3.2) and $\ln((8-4p)/p) \le \ln(8/p)$ gives, for $0 < p \le 1/2$,

$$\mathcal{L}(p) \le \frac{3p}{8}\ln\frac{1}{p} + \frac{18 + 11\ln 2}{8}p. \tag{4.6}$$

Thus every finite map obeys

$$U \ge 1 - \frac{3p}{8}\log_2\frac{1}{p} - \frac{18 + 11\ln 2}{8\ln 2}p.$$

The constant does not depend on the images, spectra, or either environment dimension. Infimizing and admitting Bell input to (1.1) preserves the inequality.

5 An admissible splitter and the matching low-noise upper

Let $E = G = \mathbb{C}^4$ and choose the four orthonormal images

$$v_0 = |0,0\rangle, \qquad v_j = \frac{|j,0\rangle + |0,j\rangle}{\sqrt{2}} \quad (1 \le j \le 3).$$

They define an isometry on E, and tracing out G defines a legal squashing map. Direct calculation gives the environment eigenvalues

$$1 - \frac{s}{2}, \quad \frac{s}{6}, \quad \frac{s}{6}, \quad \frac{s}{6}.$$

To compute the active AE spectrum, write its amplitude as

$$T = \begin{pmatrix} \alpha I_2 & \gamma X^{\mathsf{T}} & \gamma Y^{\mathsf{T}} & \gamma Z^{\mathsf{T}} \\ \gamma X^{\mathsf{T}} & 0 & 0 & 0 \\ \gamma Y^{\mathsf{T}} & 0 & 0 & 0 \\ \gamma Z^{\mathsf{T}} & 0 & 0 & 0 \end{pmatrix}, \qquad \alpha = \sqrt{(1-s)/2}, \quad \gamma = \sqrt{s/12}.$$

The map from the vacuum spin space to the three other sectors has Gram $3\gamma^2 I_2$. On its image and the vacuum spin space, T reduces to $\left(\begin{smallmatrix} \alpha & \sqrt{3}\gamma \\ \sqrt{3}\gamma & 0 \end{smallmatrix}\right) \otimes I_2$; it is zero on the orthogonal complement. Its square therefore has eigenvalues

$$\frac{1-q}{2}, \quad \frac{q}{2}, \quad \text{each twice}, \qquad q = \frac{1-\sqrt{1-s^2}}{2}.$$

Both environment and AE/AG marginals are exchange symmetric. The exact half-CMI bit cost is consequently

$$U_{\text{split}} = 1 - h_2(s/2) + h_2(q) - \frac{s}{2}\log_2 3. \tag{5.1}$$

Since $q = O(s^2)$, $h_2(q) = O(s^2 \ln(1/s))$, and $h_2(s/2) = (s/2)\log_2(1/s) + O(s)$. Equation (2.1) and the legal extension give

$$E_{\text{sq}}(\Lambda_p) \le U_{\text{split}} = 1 - \frac{3p}{8}\log_2\frac{1}{p} + O(p).$$

Together with (4.6), this proves the low-noise part of Theorem 1.1. The finite formula on $0 < p \le 1/2$, its continuous small-p expansion, and compactness away from zero give a single finite remainder constant C_0 on that full interval.

6 A faithful core and weak local isometries

Let L, R be qubits and take the core in Theorem 1.2:

$$\xi = \frac{1}{160}\begin{pmatrix} 117 & 0 & 0 & 56 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 56 & 0 & 0 & 33 \end{pmatrix}. \tag{6.1}$$

Its eigenvalues are $29/32, 1/32, 1/32, 1/32$. It is normalized, satisfies $\xi \ge I/32$, and is entangled: the odd block of its partial transpose has eigenvalue $(5-56)/160 < 0$. Also $\mathrm{Tr}(P\xi) = 131/160 > 1/2$. For an explicit purification in $G = \mathbb{C}^4$, let $v_\perp = (|00\rangle - 2|11\rangle)/\sqrt{5}$ and set

$$|\zeta\rangle = \sqrt{\frac{29}{32}}|v\rangle|0\rangle_G + \sqrt{\frac{1}{32}}(|v_\perp\rangle|1\rangle_G + |01\rangle|2\rangle_G + |10\rangle|3\rangle_G). \tag{6.2}$$

The four core vectors are orthonormal, so this is normalized and has marginal ξ.

For $0 \le r \le 1/4$, define $V_r : L \to A \otimes L$ by

$$V_r|0\rangle = |0_A 0_L\rangle, \qquad V_r|1\rangle = \sqrt{1-r}|0_A 1_L\rangle + \sqrt{r}|1_A 0_L\rangle. \tag{6.3}$$

Its columns are orthonormal. Use the same isometry on $R \to B \otimes R$ and define

$$|\psi_r\rangle = (V_r \otimes V_r \otimes I_G)|\zeta\rangle, \qquad E = LR.$$

It is a normalized pure $ABEG$ state with E, G both of dimension four. Tracing out the retained qubit in (6.3) gives D_r. Tracing out its environment gives a complementary qubit channel T_r whose excited-state survival is r and whose off-diagonal multiplier is $\sqrt{r}$.

The AB marginal is $(T_r \otimes T_r)\xi$. Its 00/11 off-diagonal entry is $7r/20$, and its populations are

$$p_{11} = \frac{33r^2}{160}, \qquad p_{01} = p_{10} = \frac{r(38-33r)}{160}, \qquad p_{00} = 1 - \frac{76r - 33r^2}{160}.$$

Hence

$$\mathrm{Tr}(P\rho_{AB}(r)) = \frac{1}{2} + \delta(r), \qquad \delta(r) = \frac{18r + 33r^2}{160}. \tag{6.4}$$

This is strictly increasing on $r \ge 0$, with $\delta(1/4) = 21/512$. Solving its quadratic gives exactly

$$r(\delta) = \frac{\sqrt{324 + 21120\delta} - 18}{66} = \frac{320\delta}{\sqrt{324 + 21120\delta} + 18}, \qquad 0 \le r(\delta) \le \frac{80}{9}\delta.$$

Thus it covers every δ in the claimed interval, including zero.

7 Entropy support, the Hessian, and uniform cubic remainders

Let $H(x, y)$ be as in (1.4). The four actual entropies of the seed are

$$S_{\mathrm{nat}}(AE) = H(0, r), \qquad S_{\mathrm{nat}}(BE) = H(r, 0),$$
$$S_{\mathrm{nat}}(E) = H(r, r), \qquad S_{\mathrm{nat}}(ABE) = S_{\mathrm{nat}}(G) = H(0, 0).$$

The first two follow by tracing the other retained qubit and removing the remaining isometry; the last follows by purity. Thus the seed's half-CMI bit cost is exactly $J(r)$. For $0 \le x, y \le 1/4$,

$$(D_x \otimes D_y)\xi \ge \frac{1}{32} D_x(I) \otimes D_y(I) \ge \frac{9}{512} I. \tag{7.1}$$

The four nonzero spectra on the isometric AE, BE supports have the same lower bound.

For completeness the matrix inside H has X-state entries

$$a' = \frac{117 + 5(x+y) + 33xy}{160}, \qquad b' = \frac{(1-y)(5+33x)}{160},$$
$$c'_{\mathrm{odd}} = \frac{(1-x)(5+33y)}{160}, \qquad d' = \frac{33(1-x)(1-y)}{160},$$
$$z' = \frac{56}{160}\sqrt{(1-x)(1-y)}.$$

Its eigenvalues are b', c'_{odd}, and

$$\lambda_\pm = \frac{a' + d' \pm \sqrt{(a'-d')^2 + 4(z')^2}}{2}.$$

On the whole square $[0,1/4]^2$, (7.1) keeps them positive, and $z' \neq 0$ keeps the discriminant positive. They are smooth on a neighborhood of this compact square. Their entropy derivatives through order three are bounded. Taylor's formula for $g(r) = H(0,r) + H(r,0) - H(r,r) - H(0,0)$ therefore gives a uniform remainder on $0 \le r \le 1/4$. Its constant and linear terms cancel.

We compute its quadratic coefficient exactly. The derivative of damping at zero is

$$\mathcal{G}(X) = |0\rangle\langle 1|\, X\, |1\rangle\langle 0| - \frac{1}{2}\{|1\rangle\langle 1|, X\}.$$

Let $A = (\mathcal{G} \otimes \mathrm{id})\xi$, $B = (\mathrm{id} \otimes \mathcal{G})\xi$, and $C = (\mathcal{G} \otimes \mathcal{G})\xi$. In units $1/160$, their diagonals are

$$A : (5, 33, -5, -33), \quad B : (5, -5, 33, -33), \quad C : (33, -33, -33, 33);$$

their $00/11$ off-diagonals are respectively $-28, -28, 14$. The differential of natural entropy gives

$$g(r) = \kappa r^2 + O(r^3), \qquad \kappa = \operatorname{Tr} C \ln \xi + \operatorname{Tr} A(D \ln_\xi)(B). \tag{7.2}$$

Indeed $\kappa = -\partial_x \partial_y H(0,0)$; the trace of C is zero.

Put $Q = I - |v\rangle\langle v|$. Since $\ln \xi = \ln(1/32) I + \ln 29\, |v\rangle\langle v|$, the required matrix elements are

$$\langle v, Cv\rangle = \frac{221}{800}, \quad \langle v, Av\rangle = \langle v, Bv\rangle = -\frac{5}{32}, \quad \langle v_\perp, Av\rangle = \langle v_\perp, Bv\rangle = \frac{1}{5},$$

$$\operatorname{Tr}(QAQBQ) = -\frac{321}{25600}.$$

In an eigenbasis of ξ, $D \ln_\xi$ multiplies entries by the logarithmic divided difference $(\ln \lambda_i - \ln \lambda_j)/(\lambda_i - \lambda_j)$, with diagonal value $1/\lambda_i$. Between $29/32$ and $1/32$ it is $(8/7)\ln 29$. Therefore the second trace in (7.2) is

$$\frac{25}{928} - \frac{321}{800} + \frac{16}{175} \ln 29.$$

Together with the first trace this gives

$$\kappa = \frac{2059}{5600} \ln 29 - \frac{2171}{5800}.$$

It is positive: $e < 3$ implies $\ln 29 > 2$, and the resulting rational lower bound is positive.

Taylor's formula and the bounded third derivative prove (1.5) on the entire interval. The inverse parameter is smooth there and satisfies

$$r(\delta) = \frac{80}{9}\delta + O(\delta^2).$$

Its derivatives are bounded on $[0, 21/512]$. Thus $r(\delta)^2 = (6400/81)\delta^2 + O(\delta^3)$ with a uniform constant, and $r(\delta)^3 \le (80/9)^3 \delta^3$. Substitution proves (1.6). Strong subadditivity gives $J(r) \ge 0$. Enlarging the finite remainder constant yields $J(r(\delta)) \le C_1 \delta^2$ on the full interval.

8 Four tetrahedral branches and the original-environment map

Let the four Bloch directions be

$$n_1 = \frac{(1,1,1)}{\sqrt{3}}, \quad n_2 = \frac{(1,-1,-1)}{\sqrt{3}}, \quad n_3 = \frac{(-1,1,-1)}{\sqrt{3}}, \quad n_4 = \frac{(-1,-1,1)}{\sqrt{3}}.$$

They satisfy $\frac{1}{4}\sum_j n_j = 0$ and $\frac{1}{4}\sum_j n_j n_j^{\mathsf{T}} = I_3/3$. For $n_j = (n_{jx}, n_{jy}, n_{jz})$ an explicit choice is

$$U_j = \begin{pmatrix} \alpha_j & -\overline{\beta_j} \\ \beta_j & \alpha_j \end{pmatrix}, \quad \alpha_j = \sqrt{\frac{1+n_{jz}}{2}}, \quad \beta_j = \frac{n_{jx}+in_{jy}}{\sqrt{2(1+n_{jz})}}.$$

It is unitary and satisfies $U_j Z U_j^\dagger = n_j \cdot \sigma$. Let $W_j = \overline{U_j} \otimes U_j$, which fixes Φ. Use independent four-dimensional flag spaces J, K and define

$$|\Omega_r\rangle = \frac{1}{2}\sum_{j=1}^{4} (W_j \otimes I_{EG})|\psi_r\rangle|j\rangle_J|j\rangle_K, \qquad E' = J \otimes LR, \quad G' = K \otimes G. \tag{8.1}$$

This is normalized and pure, with $\dim E' = \dim G' = 16$. Tracing out K kills cross-branch terms. Each branch is a local unitary of the seed, and the common J flag entropies cancel in CMI:

$$\frac{1}{2} I_2(A : B \mid E')_{\Omega_r} = J(r). \tag{8.2}$$

The seed's AB marginal has Bloch form

$$\frac{1}{4}[I + z(Z \otimes I + I \otimes Z) + s_0(X \otimes X - Y \otimes Y) + t_0 Z \otimes Z],$$

where

$$z = 1 - \frac{19r}{40}, \qquad s_0 = \frac{7r}{10}, \qquad t_0 = 1 - \frac{19r}{20} + \frac{33r^2}{40}.$$

Partial transpose on A changes the correlation tensor to $s_0 I_3 + (t_0 - s_0) e_z e_z^{\mathsf{T}}$. Under W_j, this transposed state transforms by $U_j \otimes U_j$. The tetrahedral first and second moments kill its local Bloch vectors and replace its correlation tensor by $(2s_0 + t_0)I_3/3$. Undoing partial transpose therefore gives exactly

$$\mathrm{Tr}_{E'G'} |\Omega_r\rangle\langle\Omega_r| = \rho_{\delta(r)}. \tag{8.3}$$

This also follows by comparing $(2s_0 + t_0)/3 = (1 + 4\delta(r))/3$.

For the canonical Bell basis $b_i = (I \otimes P_i)\Phi$, let $q_0 = 1/2 + \delta$, $q_i = (1/2 - \delta)/3$ for $i > 0$. All are positive on the full interval under consideration. At $r = r(\delta)$, define

$$w_i = q_i^{-1/2}(\langle b_i| \otimes I_{E'G'})|\Omega_{r(\delta)}\rangle, \quad V_\delta|i\rangle = w_i, \quad \mathcal{S}_\delta(X) = \mathrm{Tr}_{G'}\, V_\delta X V_\delta^\dagger. \tag{8.4}$$

Equation (8.3) gives $\langle w_i, w_j\rangle = \delta_{ij}$. Thus V_δ is an actual isometry from the original environment E, and $\mathcal{S}_\delta$ is CPTP with output dimension sixteen. It maps the canonical Bell purification precisely to $\Omega_{r(\delta)}$.

9 Purifier phases, coherent Pauli covariance, and every input

We verify the covariance of the pure extension, including its purifier. For a diagonal unitary $D = \mathrm{diag}(d_0, d_1)$, let

$$E_D = \overline{D}_L \otimes D_R, \quad \alpha = \overline{d_0} d_1, \quad \beta = \overline{d_1} d_0, \quad G_D = \mathrm{diag}(1, 1, \alpha^{-1}, \beta^{-1}).$$

Both $v, v_\perp$ have phase 1 under E_D; the odd core vectors have phases α, β. The explicit purifier (6.2) therefore obeys $(E_D \otimes G_D)\zeta = \zeta$. Directly on the two input basis vectors,

$$(D_{\text{retained}} \otimes D_{\text{environment}})V_r = d_0 V_r D_{\text{input}}.$$

Use its conjugate on L and ordinary version on R. The scalar factors cancel, and

$$(\overline{D}_A \otimes D_B \otimes E_D \otimes G_D)\psi_r = \psi_r. \tag{9.1}$$

For $P_i = X, Y, Z$, its Bloch rotation permutes the tetrahedral directions; write the permutation as π_i. Since the two columns of $P_i U_j$ have the eigenrays of $U_{\pi_i(j)} Z U_{\pi_i(j)}^\dagger$, there is a diagonal unitary D_{ij} such that $P_i U_j = U_{\pi_i(j)} D_{ij}$. Then

$$(P_i^{\mathsf{T}} \otimes P_i) W_j = W_{\pi_i(j)} (\overline{D_{ij}} \otimes D_{ij}).$$

Here $P_i^{\mathsf{T}} = \overline{P_i}$, including the sign for Y. Define $L_E(P_i)$ to permute branch j to $\pi_i(j)$ and apply $E_{D_{ij}}$ inside it. Define $L_G(P_i)$ similarly with $G_{D_{ij}}$. These are unitary block permutations. Applying (9.1) branch by branch in (8.1) proves

$$(P_i^{\mathsf{T}} \otimes P_i \otimes L_E(P_i) \otimes L_G(P_i))\Omega_r = \Omega_r. \tag{9.2}$$

Let $T_\delta = (I_B \otimes V_\delta) U_{p(\delta)}$. It is the joint isometry for one and the same squashing map (8.4) on every input. Since $(\mathrm{id} \otimes T_\delta)\Phi = \Omega_{r(\delta)}$, (9.2) is equivalent to

$$(P_i \otimes L_E(P_i) \otimes L_G(P_i)) T_\delta P_i = T_\delta.$$

For an input density τ, the half-CMI for any pure reference purification of τ is

$$f(\tau) = \frac{1}{2}\left[S_2(B \mid E')_{T_\delta \tau T_\delta^\dagger} + S_2(B \mid G')_{T_\delta \tau T_\delta^\dagger} \right].$$

This identity is purity's entropy duality on the reference and the three joint output systems. Conditional entropy is concave, so f is concave. The proved local covariance gives $f(P_i \tau P_i) = f(\tau)$. The qubit Pauli average is $I_2/2$, whence

$$f(\tau) \le f\left(\frac{1}{4} \sum_{i=0}^{3} P_i \tau P_i \right) = f(I_2/2) = J(r(\delta)).$$

Bell input attains equality. This proves the exact same-map maximum in Theorem 1.2. Using that single map in the inner infimum of (1.1), then taking the outer supremum, proves the boundary upper in Theorem 1.1. Together with the previous entropy section it completes all assertions of Theorem 1.2.

10 The unrestricted quadratic lower

For a Hermitian bipartite operator X, let $\|X\|_\to$ be the supremum of the classical ℓ^1 norm of the measurement outcome vector over all finite Alice-to-Bob one-way LOCC measurements. Put

$$d_\to(\rho, \mathrm{SEP}) = \inf_{\sigma \in \mathrm{SEP}} \|\rho - \sigma\|_\to.$$

We use the finite-dimensional faithfulness theorem [3, Theorem, equation (9), and Corollary 1]: for every density ω_{ABE},

$$I_{\mathrm{nat}}(A : B \mid E)_\omega \ge \frac{1}{8} d_\to(\omega_{AB}, \mathrm{SEP})^2. \tag{10.1}$$

Equivalently its half-CMI bit cost is at least $d_\to^2/(16 \ln 2)$. The one-way classical communication is from Alice to Bob. The entanglement-testing identity used in its proof is the generalized Stein equality with normalized separable alternatives [4]; it applies to singular null states as well.

The numerical witness for ρ_δ is explicit. Alice chooses one of the three Pauli axes uniformly, measures the transposed observable on her qubit, and sends the axis and outcome to Bob. Bob measures the ordinary observable and accepts equal outcomes. The binary acceptance effect is

$$M = \frac{1}{3} \sum_{i=X,Y,Z} \frac{I + P_i^{\mathsf{T}} \otimes P_i}{2} = \frac{I + 2P}{3}. \tag{10.2}$$

For any product density $\alpha \otimes \beta$, $\mathrm{Tr}(P(\alpha \otimes \beta)) = \mathrm{Tr}(\alpha^{\mathsf{T}}\beta)/2 \leq 1/2$. Convexity gives the same inequality for every separable density. Thus all separable densities have acceptance at most $2/3$, whereas

$$\mathrm{Tr}(M\rho_\delta) = \frac{2}{3} + \frac{2\delta}{3}.$$

For a binary measurement the outcome ℓ^1 distance is twice the acceptance difference. Hence

$$d_{\to}(\rho_\delta, \mathrm{SEP}) \geq \frac{4\delta}{3}.$$

Every finite squashing map on the canonical Bell purification leaves this AB marginal exactly ρ_δ. Apply (10.1) to each of these actual extensions:

$$\frac{1}{2} I_2(A : B \mid E) \geq \frac{(4\delta/3)^2}{16 \ln 2} = \frac{\delta^2}{9 \ln 2}. \tag{10.3}$$

The inequality has no conditioning-dimension restriction. Take the infimum over all finite outputs and use Bell input in the outer supremum. This proves the lower in (1.3). The already proved upper and the positive lower coefficient give its right-neighborhood $\Theta(\delta^2)$ statement.

For the core ξ, the same actual one-way effect gives $\mathrm{Tr}(M\xi) = 211/240 > 2/3$, also verifying its nonseparability directly from the original convex-hull definition of SEP.

11 A capped pressure consequence

For any positive integer D, define the natural-unit pressure

$$K_D(\ell) = \sup_{\substack{1 \leq \dim E \leq D \\ \omega_{ABE} \text{ density}}} \left\{ \ell \left(\mathrm{Tr}(P\omega_{AB}) - \frac{1}{2} \right) - I_{\mathrm{nat}}(A : B \mid E)_\omega \right\}. \tag{11.1}$$

This domain contains all normalized densities in the admitted dimensions. For $\ell \geq 0$, the Bell effect bound and strong subadditivity give every payoff at most $\ell/2$, so the supremum is finite.

Proposition 11.1. *There exist $c, \ell_0 > 0$ such that, for every $D \geq 16$ and every $0 < \ell \leq \ell_0$,*

$$K_D(\ell) \geq c\ell^2.$$

Proof. The actual E16 extension from (8.1) and (8.4) has Bell fidelity $1/2 + \delta$. The uniform entropy bounds supply a constant $C > 0$, common to $0 \leq \delta \leq 21/512$, for which its natural CMI is at most $C\delta^2$. For every $D \geq 16$ this state is admitted in (11.1); therefore $K_D(\ell) \geq \ell\delta - C\delta^2$. Choose

$$\delta = \frac{\ell}{2C}, \qquad \ell_0 = 2C\frac{21}{512}, \qquad c = \frac{1}{4C}.$$

The parameter is physical for $0 < \ell \leq \ell_0$, and substitution gives $K_D(\ell) \geq \ell^2/(4C)$. □

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